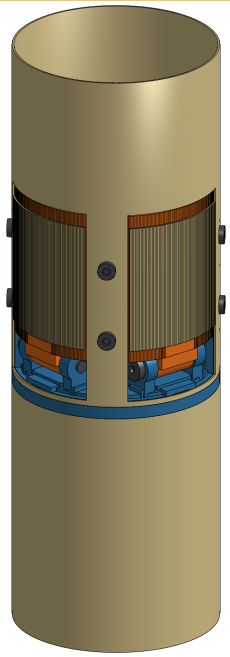
# V4 – servo on top, thicker parts, electronics bay: assembly guide

Built from the sketch you reviewed, with Option B (a little more length), the Mercury V1 altimeter, and my assumptions for the battery and regulator. The fins, cams, hinge, tube slits and rotor-mounted latch are the V3 ones; what changed is below.

Onshape: ARC BT80 Airbrake - V1 Screwed + V2 Compact (Claude build). Elements: **V4-A** closed, **V4-B** fully open, **ASM-V4** assembly. Feature Studio: module mode *4* with *V2 compact head* on.

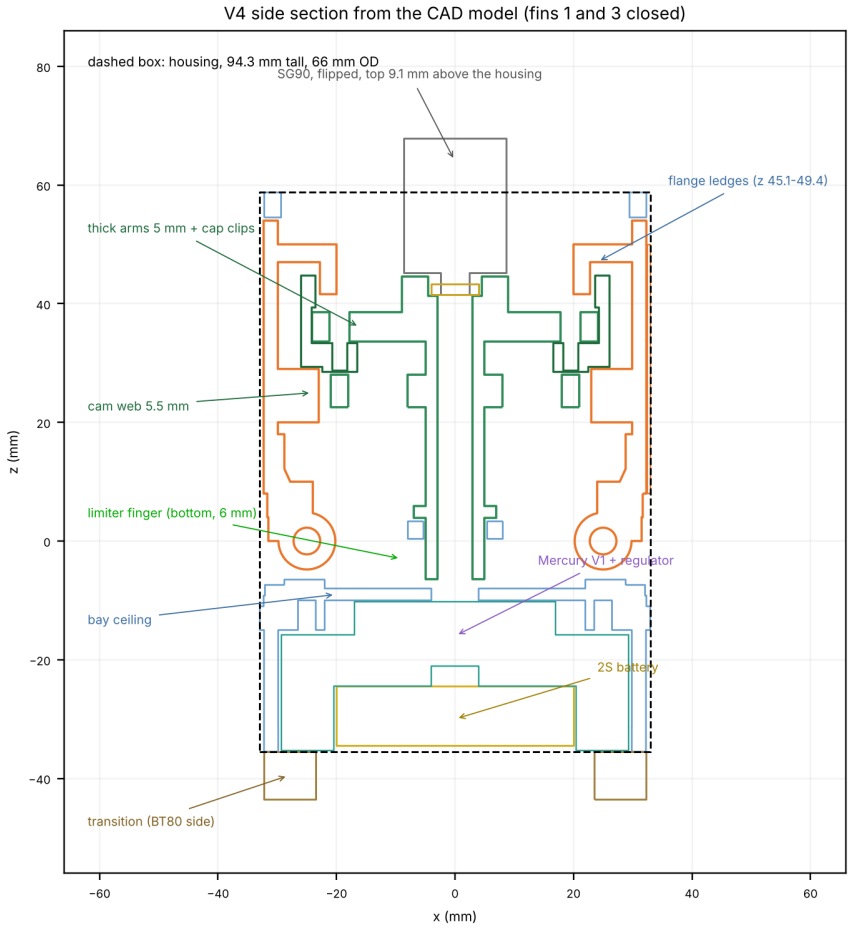
**Where it differs from the sketch:** (1) the servo top is **9.1 mm above the housing**, not flush: the catch-cap fingers sweep under the servo's flange ledges, so the ledges had to move above them, which lifts the servo; it sits in the upper BT80 tube. (2) The crown arm is 8 mm wide and 5 mm thick with no separate boss (walls beside the screw 2.3 mm, not 3.3). (3) Limiter stops are 7.6 mm wide, the cam web 5.5 mm. (4) The housing is 94.3 mm (Option B: +2.5 mm over V3), and 103.4 mm from the housing bottom to the top of the servo.



V4 closed. The upper tube, with the servo showing through its window zone, covers the head as before.

## 1 The numbers

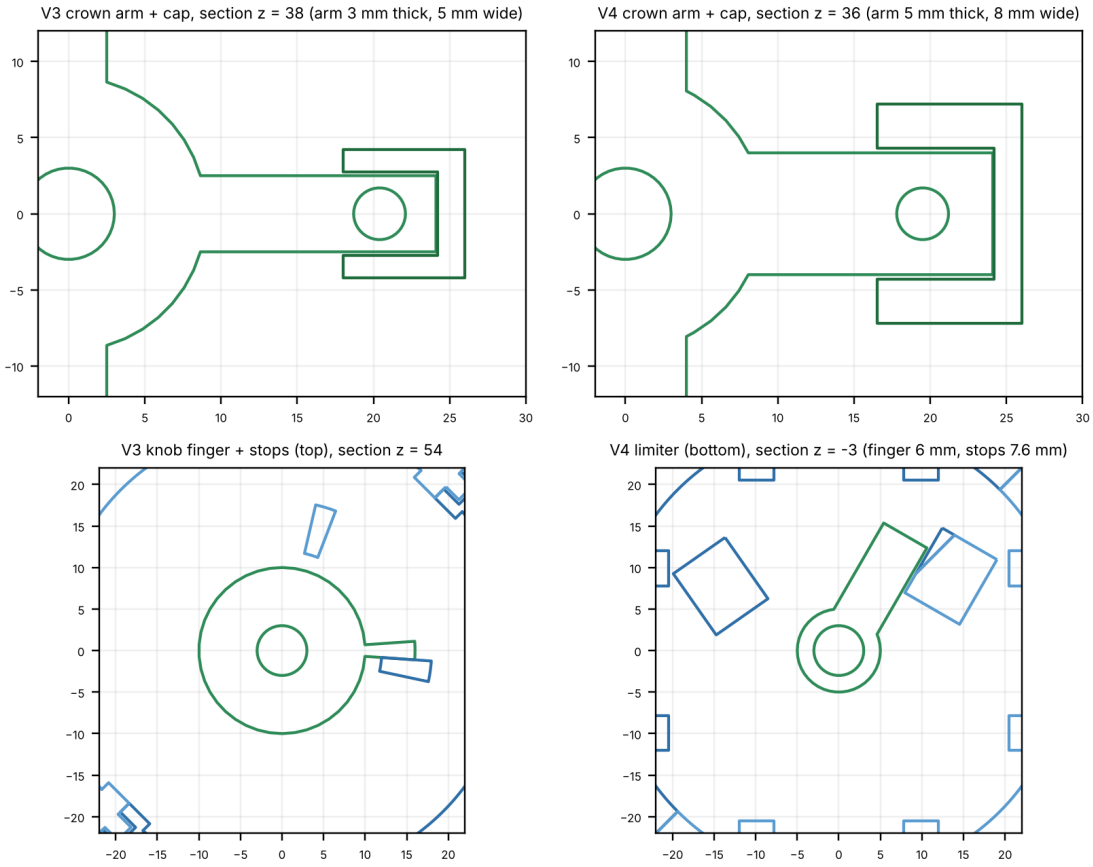
| | V3 flush | V4 |
|---|---|---|
| Printed housing | 91.8 mm | 94.3 mm (z –35.5 … 58.8) |
| Housing bottom to servo top | 91.8 mm (servo inside) | 103.4 mm (servo top 9.1 mm above the housing) |
| Printed set, solid PLA | 106 g | **143 g**: 113 g core + 16 g sled + 15 g transition ring |
| Printed part types | 5 | 7 (adds the electronics sled and the transition ring) |
| Servo | inside, shaft up, bottom flush | on top, flipped, shaft down |
| Rotation limiter | top: finger and stops about 2 mm wide | bottom: finger 6 mm, stops 7.6 mm |
| Crown arms | 5 × 3 mm, 0.8 mm walls at the screw | 8 × 5 mm, 2.3 mm walls (cap body 14 mm wide) |
| Cam web | 3 mm | 5.5 mm |
| Electronics | none | Mercury V1 + 2S LiPo + 5 V regulator in the bay |
| Tube joint at the bottom | 3 radial M3 into sleeve pads | 4 vertical M3×12 into the transition ring |




Side section from the CAD model, to scale.

## 2 What you asked for, and what I did

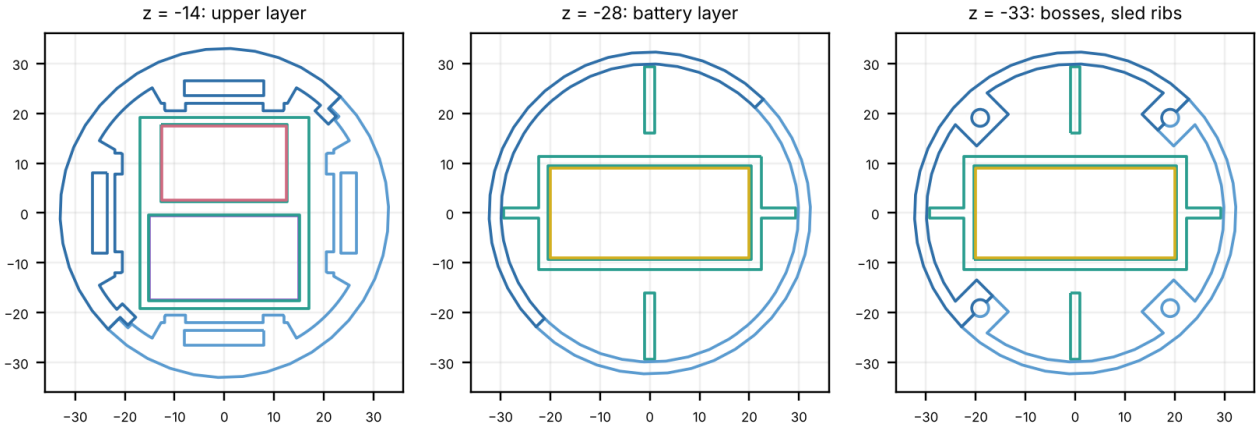
- **Thicker limiter and rotor parts.** Limiter finger 2 → 6 mm wide and 5.2 mm tall; stops 2 → 7.6 mm, 9.8 mm tall, hung from the lower spokes; arms 3 → 5 mm thick and 5 → 8 mm wide; cam web 3 → 5.5 mm. The catch caps got a 14 mm wide body to carry the wider arm.
- **Flipped.** The servo is on top with the shaft down into a pocket on the rotor hub; the limiter moved to the bottom of the rotor. The old top bearing is gone: the rotor is held by the servo horn and the lower bearing, with a collar above the bearing and the limiter finger below it.
- **Bottom end.** Four bosses on the housing take M3×12 screws that go down into the transition ring (heads in counterbores, reached from inside the bay with the sled out).
- **Bay.** A ceiling at z −10 closes the mechanism; below it a skeletal sled holds the Mercury and regulator in the upper layer and the battery in the lower layer, clamped between the ceiling and the transition.




Before and after, sections taken from the V3 and V4 models.

## 3 The electronics bay

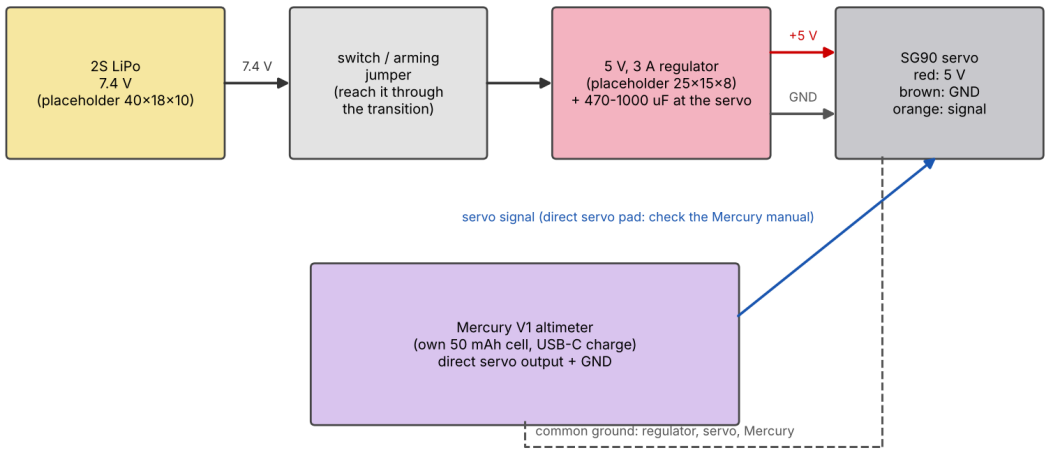
Electronics bay, plan sections of the V4 model (mm). Teal: sled, purple: Mercury V1, pink: regulator, yellow: battery




Plan sections of the V4 model. Mercury V1 is the real size (30 × 16.9 × 10.8 mm); battery and regulator are placeholders.

| Part | Size | Source |
|---|---|---|
| Mercury V1 altimeter | 30 × 16.9 × 10.8 mm, 7.2 g in case, onboard 50 mAh battery, USB-C, Wi-Fi, 2 direct servo outputs | Altimeter Cloud specification you sent |
| Servo battery | 2S LiPo, 7.4 V, about 40 × 18 × 10 mm (placeholder) | my assumption |
| Regulator | 5 V, 3 A step-down module, about 25 × 15 × 8 mm (placeholder) | my assumption |

Power and signal (sketch). The servo is powered by the regulator, not by the Mercury; the Mercury only sends the position signal.




Power and signal. The Mercury has its own battery and drives the servo through a direct servo output; the servo is powered separately.

## 4 Why the servo stands 9.1 mm above the housing

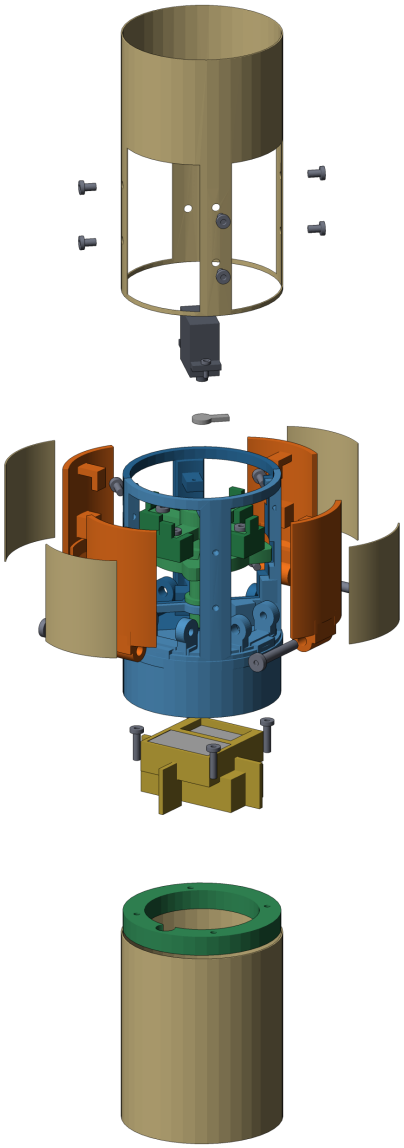
The four catch-cap fingers sweep 65° around the axis and pass under every azimuth, so a servo ledge that sits below the fingers is hit between 30° and 60° of rotor travel (found in the interference sweep, 63 mm³ at 45°). Raising the ledges above the fingers fixes that for good, and lifts the servo 6 mm; then I brought the housing top back down to 58.8 mm because it no longer needs to enclose the servo. Alternatives, if you do not want the servo above the ring: lower the whole latch (needs a redesign of the fin lug), or hold the servo by a yoke at its top end (adds a lid and moves the support above the nose-side ring).

## 5 Hardware

| Ref in CAD | Fastener | Qty | Joint | Driver |
|---|---|---|---|---|
| HW seam screw 1-4 | M3 × 8 socket-head cap | 4 | Seam, z = 8 and z = 44 | 2.5 mm hex |
| HW L1-L4 cap screw | M3 × 10 button head | 4 | Rotor arm → catch cap boss | 2.0 mm hex |
| HW H1-H4 hinge screw | M4 × 25 button head | 4 | Fin hinge | 2.5 mm hex |
| HW top-tube support screw 1-8 | M3 × 5 button head | 8 | Upper BT80 tube → strips | 2.0 mm hex |
| HW servo flange screw 1-2 | M2 × 6 button head, self-tapping | 2 | Servo tabs → ledges (from above) | 1.5 mm hex or PH0 |
| HW horn screw | M2 × 6 (supplied with the servo) | 1 | Horn → spline, from below through the rotor bore | PH0 / 1.5 mm hex |
| HW transition screw 1-4 | M3 × 12 button head | 4 | Module bosses → transition ring (from the bay side) | 2.0 mm hex |

**Totals:** 27 screws: M3×5 × 8, M3×8 × 4, M3×10 × 4, M3×12 × 4, M4×25 × 4, M2×6 × 3.

Print list (STLs in `onshape/exports/v4/print`): housing halves C1 and C2, fin ×4, catch cap ×4, rotor, **electronics sled**, **transition ring**. The sled and the transition ring are new; the transition ring is yours to bond into the lower BT80 tube.



Exploded V4. Top to bottom: upper tube, servo with horn, housing with rotor and fins, electronics sled with its parts, transition ring, lower tube.

## 6 Assembly sequence

- Hinge screws, catch caps (M3×10 into the arm bosses), rotor into one housing half, fins, second half, four seam screws, as in V3.
- Set the servo to its 0° position, press the horn on so that its arm lies in the slot at 45° to the crown arms, and lower the servo with its horn into the pocket on the rotor hub from above.
- Screw the servo tabs to the two ledges with the M2×6 screws from above. Put the horn screw in from below through the rotor bore and the ceiling hole (R3 bore, small driver).
- Fill the sled: Mercury and regulator in the top pockets, battery in the bottom pocket; run the servo wires down the hollow rotor and through the ceiling hole to the bay; slide the sled up into the bay.
- Bond the transition ring into the lower BT80 tube, set the module on it, and fit the four M3×12 screws down through the bosses from inside the bay (before the sled goes in, or with the sled out).
- Glue the tube cut-outs to the fins and slide the upper tube on with its eight M3×5 screws, as in V2 and V3. The top of the servo sits inside this tube.

## 7 Checks run on this model

| Check | Result |
|---|---|
| V4-A closed and V4-B open (90°, 65°) | 51 bodies, all watertight; no printed-part overlap; the only overlaps are the servo spline inside the horn and the reference tube touching the housing (as in V2/V3) |
| Rotor sweep –1.5° … 66.5° | clear of both housing halves, the servo, the sled and the transition ring at every step |
| Limiter | engages at –3° and 67°: the 0–65° travel has about 1.5° of margin at each end |
| Latch and cam | latch holds from about 1.5° of fin opening and frees by 15° of rotor travel; cams first touch at about 26° (same as V3) |
| Fin + cut-out swept 0.5 … 90° | clear of the upper tube and the housing |
| Seam screws, servo screws, horn screw, transition screws | a hex key reaches all of them (servo screws from above, horn screw through the bore, transition screws from inside the bay) |
| V1, V2, V3 rebuilt | volumes identical to before: nothing in them changed |
| ASM-V4 mates | 21 features: fin groups, four hinges, rotor revolute: all solve |

**Not verified, and what I need from you:** (1) your servo and horn dimensions (the ledge pilots and the horn pocket must match). (2) The Mercury's servo output: which pad, and whether its 3.3 V signal level drives the SG90 (check the manual). (3) Battery and regulator sizes are placeholders. (4) **Pressure:** the Mercury reads static pressure, so the bay needs vent holes to the outside (not modelled): about three 3 mm holes through the lower tube at the Mercury level. (5) The servo cable must come down through the hollow rotor (3 wires beside the 4 mm horn-screw head in a 6 mm bore): tight, to be tried on a print. (6) USB-C access to the Mercury is with the sled out. (7) Latch and horn pocket not tested on a print; ground-impact FEA not run (the ledges, limiter stops and the 1.4 mm sled walls are the first places to look).